Kunche Mahani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.01
Narayana Junior College,Nellore	Class XII (Board of Intermediate Education: A.P., Tadepalli)	2023	97.80%
Mythilee English Medium High School, Nellore	Class X (Board of Secondary Education, A.P.)	2021	100.00%

SCHOLASTIC ACHIEVEMENTS

- Cleared JEE MAINS ,JEE ADVANCED and qualified AP EAMCET 2023 .
- Secured admission into IIT Hyderabad through JEE Advanced.

INTERNSHIP EXPERIENCE

VH Mobile Metrics Pvt. Ltd. | Software Trainee

Jun 2026 - Jul 2026 | ***AeroInspect AI***

- Engineered backend-frontend integration using FastAPI & PostgreSQL to process real-time YOLO detections, **automating PPE violation tracking** and reducing reliance on manual inspections.
- Architected **worker registration** and accountability workflows with persistent profiles and violation records, enabling **worker-level compliance tracking** and auditable safety histories.
- Developed an **LLM-powered rule-extraction pipeline** converting safety manuals into structured rules for automated compliance checks, risk alerts and reporting; fixed PDF truncation that excluded **~71% of source content** from analysis.

PROJECTS

Notification Framework -- Scalable Multi-Channel Notification Platform

- Engineered a unified REST API supporting **4 notification channels -- Email, WhatsApp, SMS and Push** -- through a single interface for multi-channel delivery.
- Designed an asynchronous queue-based architecture using Celery and RabbitMQ with retry handling and DLQ support, enabling reliable **notification delivery** at scale.
- Implemented rate limiting, user preferences and DND scheduling with a React admin dashboard for **queue monitoring, webhook tracking and delivery analytics**.

Load Balancing Algorithm Simulator

- Benchmarked **6 load-balancing algorithms** -- Round Robin, Weighted Round Robin, Random, IP Hash, Least Connection and Weighted Least Connection -- across varying server configurations.
- Developed a **custom Mininet network topology** with multithreaded Python clients to simulate concurrent workloads and measure response-time performance.
- Analyzed **performance trade-offs** across 3, 5 and 7-server configurations under increasing client loads.

Multi-Client TCP Quiz System

- Developed a multi-client TCP quiz server using multithreading to support **concurrent players** with parallel request handling and synchronized quiz sessions.
- Designed a custom client-server communication protocol for **reliable message exchange and synchronization** across multiple concurrent clients.
- Integrated LLM-based question generation and evaluated **system latency under concurrent loads**.

SKILLS

Programming Languages: C, C++, Python, SQL

Core & Problem Solving: Data Structures & Algorithms, Problem Solving (DSA), Software Engineering, OOP

Backend & Databases: FastAPI, SQLAlchemy, PostgreSQL, REST APIs

Python & ML: NumPy, Pandas, PyTorch, Scikit-learn, Matplotlib, Machine Learning, Deep Learning

Web & AI Integration: HTML, CSS, React, Tailwind CSS, LLM APIs

Tools & Platforms: Git, GitHub, Docker, Postman, VS Code, Linux

Soft Skills: Leadership, Analytical Thinking, Effective Communication, Adaptability, Curiosity.

RELEVANT COURSES

Relevant Coursework: Probability & Random Variables, Optimization, Data Structures & Algorithms, Numerical Methods, Linear Algebra, Statistics, Computer Networks, Machine Learning, Database Management Systems, Operating Systems.

POSITIONS OF RESPONSIBILITY

Events Coordinator &Hospitality Head , Milan'24 & Milan'25: Managed hospitality and Inter-Hostel Championship events across sports, cultural and technical competitions.

Publicity Coordinator, Elan & nVision'25: Led fest publicity, coordinating with professors and HODs across colleges to promote workshops, events and campus performances

Sunshine Buddy: Mentored first-year students, supporting their transition to campus life and academics.

EXTRACURRICULARS & INTERESTS

- **Sports:** Member of the National Sports Organisation (NSO), IIT Hyderabad; represented IITH at the **Inter-IIT Sports Meet 2024 in Athletics** and actively participated in **Chess and Football**, developing **teamwork and discipline**.
- **Interests:** Passionate about **fashion and dance**, actively participating in dance performances and **organizing and conducting events** at IIT Hyderabad.